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| **3**<br><br>**Further matrix algebra** | 3.1 | **Eigenvalues and eigenvectors of 2 × 2** and 3 × 3 **matrices.** | **Understand the term *characteristic equation* for a 2 × 2 matrix.**<br><br>**Repeated eigenvalues and complex eigenvalues.**<br><br>**Normalised vectors may be required.** |

| Topics | What students need to learn: | | |
| --- | --- | --- | --- |
| | Content | | Guidance |
| **3**<br>**Further matrix algebra**<br>*continued* | 3.2 | **Reduction of matrices to diagonal form.** | **Students should be able to find a matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{AP}$ is diagonal**<br>**Symmetric matrices and orthogonal diagonalisation.** |
| | 3.3 | **The use of the Cayley-Hamilton theorem.** | **Students should understand and be able to use the fact that, every $2\times2$** or **$3\times3$ matrix satisfies its own characteristic equation.** |